



Tree Physiology 35, 1343–1355
doi:10.1093/treephys/tpv075



Research paper

Root–soil air gap and resistance to water flow at the soil–root interface of *Robinia pseudoacacia*

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Received December 3, 2014; accepted July 21, 2015; published online September 9, 2015; handling Editor Ram Oren

During periods of water deficit, growing roots may shrink, retaining only partial contact with the soil. In this study, known mathematical models were used to calculate the root–soil air gap and water flow resistance at the soil–root interface, respectively, of *Robinia pseudoacacia* L. under different water conditions. Using a digital camera, the root–soil air gap of *R. pseudoacacia* was investigated in a root growth chamber; this root–soil air gap and the model-inferred water flow resistance at the soil–root interface were compared with predictions based on a separate outdoor experiment. The results indicated progressively greater root shrinkage and loss of root–soil contact with decreasing soil water potential. The average widths of the root–soil air gap for *R. pseudoacacia* in open fields and in the root growth chamber were 0.24 and 0.39 mm, respectively. The resistance to water flow at the soil–root interface in both environments increased with decreasing soil water potential. Stepwise regression analysis demonstrated that soil water potential and soil temperature were the best predictors of variation in the root–soil air gap. A combination of soil water potential, soil temperature, root–air water potential difference and soil–root water potential difference best predicted the resistance to water flow at the soil–root interface.

Keywords: *Robinia pseudoacacia*, root–soil air gap, soil–root interface, water flow resistance at the soil–root interface.

Introduction

Plant roots are important organs for water and nutrient uptake, the synthesis of growth regulators and carbohydrate storage in the soil–plant–atmosphere continuum (Heeraman et al. 1997). Root contact with the soil is essential for water and nutrient absorption by plants (North and Nobel 1997b). Soil properties (e.g., water content, bulk density, volume, degree of compaction and average particle size) along with root properties (e.g., root diameter and relative hydration) can influence the extent of root–soil contact (Tinker 1976, Nye 1994, North and Nobel 1997b). Root–soil contact may be poor for plants in sandy, shallow and shifting soils, which are loose in structure and easily infiltrated; this poor contact can lead to air gaps between the roots and the soil (North and Nobel 1997a). The degree of

root–soil contact for plants growing in structured soil varies among root segments, ranging from 0 to 100% of the root perimeter (Van Noordwijk et al. 1992). After losing contact with the soil, roots take up water via vapour diffusion (Cowan and Milthorpe 1968, Nobel and Cui 1992a); however, gas-phase water flow is much slower than liquid-phase flow for a wide range of water potentials (Carminati et al. 2009). For these reasons, incomplete root–soil contact due to porous or fissured soil structures or root or soil shrinkage can reduce the uptake of water and nutrients (Faiz and Weatherley 1982, Veen et al. 1992, North and Nobel 1997b). Alternatively, considerable increases in root–soil contact improve the uptake efficiency of nutrients and water (Willigen and Noordwijk 1987). Although roots growing in heavily compacted or waterlogged soil may

have considerable soil contact, problems with root gas exchange may be caused by the absence of air spaces between roots and soil particles (Veen et al. 1992).

Water flow in soil–plant systems is a function of physical soil properties (e.g., soil water content), plant physiology (e.g., effective root length) and meteorological factors (e.g., atmospheric evaporation) (Gardner 1960, Tomar and Ghildyal 1975, Green et al. 2003). Water uptake by plant roots in unsaturated soil is controlled by the water potential gradients across three components of the soil–root system: the soil, the root–soil air gap and the root (Gardner 1960, Nobel 1991, Huang and Kang 1997, Green et al. 2003). The soil water potential determines which component is the limiting factor in water flow in the soil–root system (Huang and Kang 1997, Guo and Li 1999). Water movement into roots from wet soil in the range of 0 to -0.13 MPa is limited by root resistance (Huang and Kang 1997, Zarebanadkouki et al. 2012). As soil becomes dry, water movement is limited mainly by resistance to water flow at the soil–root interface. As the soil continues to dry, water movement is dominated by soil resistance. Gardner and Ehlig (1962) compared the total resistance of water movement from the soil across an air gap to a root with that predicted for the soil alone; they found that when soil suction exceeded 1 or 2 bars, soil resistance became the limiting factor. Newman (1969a), however, reported that soil resistance remains lower than plant resistance until the permanent wilting point is neared or passed. Moreover, in other studies soil resistance remained lower than plant resistance over the entire range of soil water availability (Blizzard and Boyer 1980, Ruggiero et al. 1999). Kang and Zhang (1997) indicated that root shrinkage and water flow resistance at the soil–root interface differ based on the plant type for the same soil water potential, while Guo and Li (1999) suggested that the water flow resistance at the soil–root interface increases with growth time and decreases with soil moisture. Water uptake by plant roots is determined by the radial resistance from the root surface to the root xylem along with the axial resistance to movement along the root xylem (Frensch and Steudle 1989, Zwieniecki et al. 2003). The species, growth time and soil water content all affect the axial and radial pathways for water movement (Zwieniecki et al. 2003), and radial resistance has been shown to be more important in limiting water uptake than the axial resistance (Frensch and Steudle 1989). Xylem bubbles (embolisms), which are induced by cavitation events due to water stress or freezing, can greatly reduce the hydraulic conductivity of xylem tissue (Tyree and Yang 1992).

A variety of water flow models have been proposed for the soil–root system (van den Honert 1948, Cowan 1965, Molz and Remson 1970, Feddes et al. 1976, Molz 1976, 1981, Ojha and Rai 1996, Luo et al. 2000, Liu et al. 2001, Zhuang et al. 2001). By applying an analogue of Ohm's law to water movement in plants, van den Honert (1948) demonstrated that in steady-state conditions, the model can be applied simultaneously to water flow in all parts of the soil–plant–atmosphere system; the

steady-state flux of water through the soil–plant–atmosphere continuum is proportional to the water potential gradient and inversely proportional to the flow resistance (Elfvig et al. 1972). This model has been used by Lynn and Carlson (1990), Braud et al. (1995), Ridder and Schayes (1997) and Couvreur et al. (2012), among others. Cowan (1965), Molz and Remson (1970), Feddes et al. (1976), Molz (1976) and de Jong van Lier et al. (2008) developed microscopic and macroscopic models describing the flow of soil moisture towards plant roots that absorb water at periodically varying rates. Molz (1981), Ojha and Rai (1996), Doussan et al. (1998) and Luo et al. (2000) proposed mathematical models describing water movement in root-containing soil and water flow in living plant tissue. Doussan et al. (2006) and Javaux et al. (2008) presented models describing both soil and plant processes involved in water uptake on the scale of the entire root system. Despite the large number of models, few attempts have been made to study the liquid–gas–liquid phase change processes involved in water flow at the soil–root interface, and the diffusion of water vapour across the root–soil air gap has also received little quantitative attention (Nobel and Cui 1992b).

The non-destructive and non-invasive visualization and quantification of roots is essential to monitor continuously the spatial and temporal variations in the physical and chemical parameters of the root–soil system (Kuchenbuch and Ingram 2002, Alony and Linker 2013). Root growth dynamics and structure along with rhizosphere characteristics have been measured using X-ray computed tomography (Hamza et al. 2001, Gregory et al. 2003, Perret et al. 2007, Curtis and Claassen 2008), nuclear magnetic resonance (Brown et al. 1991, van der Weerd et al. 2001), and charge-coupled device cameras (Assmus et al. 1997, Iijima et al. 2003, Urashima et al. 2005, Alony and Linker 2013). In this study, the mathematical models of van den Honert (1948) and Huang and Kang (1997) were used to calculate the root–soil air gap and the resistance to water flow at the soil–root interface, respectively, of *Robinia pseudoacacia* L. when water is transported from the soil to a root across an air gap under different soil water conditions. The root–soil air gap of *R. pseudoacacia* in a root growth chamber was investigated using a digital camera, and the measured air gap along with the model-inferred water flow resistance at the soil–root interface were compared with predictions based on a separate outdoor experiment.

Materials and methods

Models of water flow

Van den Honert (1948) determined that in steady-state conditions, the same quantity of water per hour passes successively through the root, the stem, the plant leaves and the air around the leaf. Based on this fact, he proposed a water flow model for the soil–plant–atmosphere system under steady-state conditions based on Ohm's law:

$$q = \frac{\Psi_r - \Psi_s}{R_r} = \frac{\Psi_x - \Psi_r}{R_x} = \frac{\Psi_l - \Psi_x}{R_l} = \frac{\Psi_a - \Psi_l}{R_a}, \quad (1)$$

where q is the rate of flow through the system ($\text{g m}^{-2} \text{s}^{-1}$), Ψ_s , Ψ_r , Ψ_x , Ψ_l and Ψ_a are the water potentials in the soil, root, stem, plant leaves and air, respectively (MPa), and R_r , R_x , R_l and R_a are the resistances of the root, stem, plant leaves and air, respectively (MPa s m^{-1}).

At steady state, the rate of water flow at the root surface (q ; $\text{g m}^{-2} \text{s}^{-1}$) is constant across the three components of the root–soil system. For the entry or exit of water from roots, q can be expressed as follows

$$q = \frac{\Psi_s - \Psi_{rs}}{R_s} = \frac{\Psi_{rs} - \Psi_\lambda}{R_r} = \frac{\Psi_\lambda - \Psi_l}{R_\lambda + R_l} = \frac{\Psi_l - \Psi_a}{R_l + R_a}, \quad (2)$$

where Ψ_{rs} and Ψ_λ are the water potentials at the soil–root interface and the stem xylem, respectively (MPa), and R_s and R_λ are the resistances of the soil and the stem xylem, respectively (MPa s m^{-1}).

According to Eq. (2), the resistance to water flow from the soil around the root to the root xylem (R_{rs}) consists of two parts: (i) the resistance to water flow at the soil–root interface (R'_{rs}) from the soil around the root to the root surface and (ii) the radial resistance from the root surface to the root xylem (R_r).

$$R_{rs} = \frac{\Psi_s - \Psi_r}{J_v} = R'_{rs} + R_r, \quad (3)$$

$$R'_{rs} = \frac{\Psi_s - \Psi_{ir}}{J_v}, \quad (4)$$

$$R_r = \frac{\Psi_{ir} - \Psi_r}{J_v}, \quad (5)$$

where Ψ_{ir} is the water potential at the root surface (MPa), and J_v is the volumetric flux density of water ($\text{m}^3 \text{m}^{-2} \text{s}^{-1}$ or m s^{-1}).

Water can be transported across the soil–root interface in the liquid phase, gas phase or gas–liquid mixed phase; these different states affect the resistance to water flow. Each of these three possible water flow phases is described below:

- (i) *Liquid phase*: when the soil water content reaches saturation or supersaturation, the root–soil air gap is completely filled with water, and the root directly contacts the water. In this case, water is transported across the soil–root interface in the liquid phase. The water flow resistance (R_{rs}) consists mostly of the root radial resistance from the root surface to the root xylem (R_r), and $R'_{rs} \approx 0$, $\Psi_{ir} = 0$.

$$R_{rs} = R_r = (0 - \Psi_r)/J_v. \quad (6)$$

The values of Ψ_r and J_v can be measured experimentally.

- (ii) *Gas phase*: when the soil is exposed to drought, no liquid water is present in the root–soil air gap. In this case, water is transported across the soil–root interface to the root

xylem in the gas phase (Nobel and Cui 1992a), and $R_{rs} = (\Psi_s - \Psi_r)/J_v = R'_{rs} + R_r$.

The model of Huang and Kang (1997) is given by

$$R_{sr,i} = \frac{R \cdot T \cdot D_{wv}}{V_w \cdot \Delta X_{sr}}, \quad (7)$$

and Fick's first law is written as (Nobel and Cui 1992a, Huang and Kang 1997):

$$J_{wv} = D_{wv} \cdot \Delta P_{wv} / (R \cdot T \cdot \Delta X_{sr}). \quad (8)$$

Substitution of Eqs (7) and (8) into Eq. (3) leads to

$$R'_{rs} = \frac{\Psi_s - \Psi_{ir}}{J_v} = \frac{R \cdot T \cdot \Delta X_{sr}}{V_w \cdot D_{wv}}, \quad (9)$$

where $R_{sr,i}$ is the resistance to water flow at the soil–root interface of the i th soil layer (MPa s m^{-1}), R is the gas constant ($\text{MPa m}^3 \text{mol}^{-1} \text{K}^{-1}$), T is the absolute temperature (K), V_w is the partial molar volume of water ($\text{m}^3 \text{mol}^{-1}$), D_{wv} is the water vapour diffusion coefficient ($\text{m}^2 \text{s}^{-1}$), ΔX_{sr} is the root–soil air gap (m), which is the average radial distance from the soil around the root to the root surface, J_{wv} is the molar flux density of water vapour ($\text{mol m}^{-2} \text{s}^{-1}$), which can be multiplied by the partial molar volume of water to give the volumetric flux density of water (J_v , $\text{m}^3 \text{m}^{-2} \text{s}^{-1}$), and ΔP_{wv} is the water vapour pressure deficit at the soil–root interface, $\Delta P_{wv} = \Psi_s - \Psi_{ir}$ (Nobel and Cui 1992a, Huang and Kang 1997).

The value of R_r is expressed as Eq. (5). The values of ΔX_{sr} and J_v can be experimentally measured. The values of D_{wv} , V_w , R and T can be estimated from related materials.

By combining Eqs (3), (5) and (9), the value of Ψ_{ir} is obtained as

$$R_{rs} = \frac{\Psi_s - \Psi_r}{J_v} = \frac{R \cdot T \cdot \Delta X_{sr}}{V_w \cdot D_{wv}} + \frac{\Psi_{ir} - \Psi_r}{J_v}, \quad (10)$$

$$\Psi_{ir} = \Psi_s - \frac{J_v \cdot R \cdot T \cdot \Delta X_{sr}}{V_w \cdot D_{wv}}. \quad (11)$$

- (iii) *Gas–liquid mixed phase*: water can also be transported across the soil–root interface to the root xylem as a gas–liquid mixed phase, where $J_v = J_{lv} + J_{gv}$ and $R_r = (\Psi_{ir} - \Psi_r)/J_v$. R'_{rs} must consider the proportion of water transferred in a two-state liquid–gas mixture, which changes with the degree of soil aridity. In this case, we added a parameter $[\theta(c)]$ describing the proportion of water transferred in the liquid/gas phase:

$$\theta(c) = \frac{C_f - C_t}{C_f}, \quad (12)$$

$$J_{gv} = \theta(c) \cdot J_v, \quad (13)$$

$$J_{lv} = (1 - \theta(c)) \cdot J_v, \quad (14)$$

$$\begin{aligned}
 R'_{rs} &= (\psi_s - \psi_{ir})/J_v \\
 &= (\psi_s - \psi_{ir})/(J_{lv} + J_{gv}) \\
 &= [(\psi_s - \psi_{ir}) \cdot J_{lv}^{-1} \cdot J_v + (\psi_s - \psi_{ir}) \cdot J_{gv}^{-1} \cdot J_{gv}]/J_v.
 \end{aligned} \quad (15)$$

In Eqs (12)–(15), $\theta(c)$ is the degree of soil water deficit, $0 < \theta(c) \leq 1$ (dimensionless), C_t is the soil water content (g g^{-1}), C_f is the saturated soil water content (g g^{-1}) and $(\psi_s - \psi_{ir}) \cdot J_v^{-1}$ is the resistance to liquid water flow at the soil–root interface. At this point, liquid water is first vapourized at the soil–root interface, diffused, condensed at the root surface and then absorbed by roots (Huang and Kang 1997); thus, $\psi_{ir} = 0$, $(\psi_s - \psi_{ir}) \cdot J_v^{-1} \approx 0$ and $(\psi_s - \psi_{ir}) \cdot J_{gv}^{-1}$ are the resistances to gaseous water flow at the soil–root interface $(\psi_s - \psi_{ir}) \cdot J_{gv}^{-1} = (R \cdot T \cdot \Delta X_{sr})/V_w \cdot D_{wv}$. Thus,

$$\begin{aligned}
 R'_{rs} &= [0 \cdot J_{lv} + R \cdot T \cdot \Delta X_{sr} \cdot V_w^{-1} \cdot D_{wv}^{-1} \cdot J_{gv}]/J_v \\
 &= R \cdot T \cdot \Delta X_{sr} \cdot V_w^{-1} \cdot D_{wv}^{-1} \cdot J_{gv}/J_v.
 \end{aligned} \quad (16)$$

The substitution of Eq. (16) into Eq. (13) leads to

$$R'_{rs} = \theta(c) \cdot R \cdot T \cdot \Delta X_{sr} \cdot V_w^{-1} \cdot D_{wv}^{-1}. \quad (17)$$

The substitution of Eq. (17) into Eqs (3) and (5) gives

$$R_{rs} = \theta(c) \cdot R \cdot T \cdot \Delta X_{sr} \cdot V_w^{-1} \cdot D_{wv}^{-1} + \frac{\psi_{ir} - \psi_r}{J_v}. \quad (18)$$

The values of ψ_s , ψ_r , ΔX_{sr} and J_v can be measured experimentally. The values of D_{wv} , V_w , R and T can be estimated from related materials.

By combining Eqs (3), (5), (17) and (18), the value of ψ_{ir} is obtained as

$$\psi_{ir} = \psi_s - \frac{J_v \cdot \theta(c) \cdot R \cdot T \cdot \Delta X_{sr}}{V_w \cdot D_{wv}}. \quad (19)$$

Site description

This study was conducted at the Hilly Ecosystem Experimental Station, China, which is run by the Chinese Academy of Sciences (114°15'50"E, 37°52'44"N). The station is situated on Taihang Mountain, at ~50 km southwest of the city of Shijiazhuang, the capital of Hebei Province in northern China, at an elevation ranging from 247 to 1040 m above sea level. The climate is generally dry from October to May and rainy from June to September. Annual precipitation and evaporation average 560 and 1934 mm, respectively. The long-term average annual temperature is 13 °C. Soils in the area are generally classified as mountainous cinnamon soils (Ustalf). These soils possess a pH of 7.7, an organic matter content of 37,348 mg kg⁻¹, a total N of 1877 mg kg⁻¹ and a total P of 579 mg kg⁻¹ (Liu et al. 2010). A detailed site description can be found in Liu et al. (2014).

Vegetation coverage at the study site is variable and includes planted vegetation, shrubs and herbs (Liu et al. 2012). *Robinia*

pseudoacacia is widely used in afforestation, shelterbelt construction and various ecological projects in China. Therefore, *R. pseudoacacia* forests are the most common forests in Taihang Mountain. *Robinia pseudoacacia* was selected as a representative tree species to study the movement of water from the soil to a root across an air gap under different soil water conditions.

Outdoor observation

Robinia pseudoacacia planted in open fields was used to obtain the model parameters (i.e., soil water potential, leaf water potential, single root water potential, total root water potential, transpiration rate, air temperature and soil temperature) for calculating the root–soil air gap and the water flow resistance at the soil–root interface. Soil water potential and single root water potential were used to calculate the root–soil air gap [Eqs (11) and (19)], while plant transpiration rate, total root water potential, leaf water potential and root–soil air gap were used to calculate the resistance to water flow at the soil–root interface [Eqs (9) and (17)]. Two-year-old *R. pseudoacacia* plants (~2 m in height and 3 cm in basal diameter) were planted in the open fields of the Hilly Ecosystem Experimental Station in the March of 2011 with spacings of 1 × 1 m. The plants were free of fertilizers, insecticides and shields, and were watered to saturation at the beginning of each month from June to October 2011. Three plants were randomly selected to monitor the water potential and transpiration rate after watering. To facilitate the simulation of root water potential, mature leaves close to the ground were covered in foil to balance the leaf water potential towards the trunks near the ground.

Soil water potential at soil depths of 0–60 cm (where the roots of 2-year-old *R. pseudoacacia* are concentrated), leaf water potential at a height of 1.5 m, single root water potential at a soil depth of 0–60 cm, and total root water potential simulated from the water potential of leaves on near-ground trunks were measured using a WP4 Dew-point Potentiometer (Decagon Devices Inc., Pullman, WA, USA) with a 0.5 m KCl solution at 2-h intervals from 08:00 to 18:00 h (Curtis and Claassen 2008, Gubiani et al. 2013). Soil near the tree trunks was collected at four depths (0–10, 10–20, 20–40 and 40–60 cm) at 2-h intervals from 08:00 to 18:00 h using a soil sampler. Roots in each soil layer were collected and sealed immediately in separate small plastic bags containing moist paper towels and stored briefly in a cooler. At the same time, three fully expanded and intact leaves in direct sunlight from each tree at near-ground and 1.5 m above-ground were cut and sealed immediately in separate small plastic bags and stored briefly in a cooler. Root and leaf samples were cut into pieces, and the soil, root and leaf samples were transferred to the WP4 sample cup, sealed with a lid and left on the WP4 for ~5 min to equilibrate the temperature of the sample to that of the chamber. Subsequently, the average soil water potential and single-root water potential at a soil depth of 0–60 cm, the average leaf water potential at a height of 1.5 m and the average total root water potential were obtained.

The plant transpiration rates from 08:00 to 18:00 h were measured using a Flow-32 system (Dynamax Inc., Houston, TX, USA). The gauges were installed at a height >0.2 m above the ground surface. Due to a malfunction of sensors, they were disconnected in September. The gauge output was monitored every 60 s and recorded at 15-min intervals with a CR1000 data logger (Campbell Scientific, Logan, UT, USA) (Ding et al. 2013). Further detail on the principle and installation procedure can be found in Trambouze and Voltz (2001). The diameter at breast height (1.3 m) was measured using a caliper.

Soil temperatures at depths of 5, 15 and 25 cm near the trees were measured with a standard soil thermometer at 2-h intervals from 08:00 to 18:00 h. Soil bulk density at a soil depth of 0–60 cm, field capacity and saturation moisture capacity were measured using the cutting ring method (100 cm³ per sample). Solar radiation, air temperature, air humidity, rainfall and wind speed at 2 m above ground were measured by a standard automatic weather station at the Hilly Ecosystem Experimental Station.

Indoor observation

Minirhizotrons were buried in the open fields to observe changes in the root system for comparison with the model results in order to improve the practical application value of the model. Unfortunately, many rocks in the soil impede either the ability of roots to grow along the minirhizotrons or our ability to observe them (Hendrick and Pregitzer 1996); therefore, we conducted a complementary indoor observation experiment. In this experiment, a 1-year-old *R. pseudoacacia* planted in a root growth chamber was used to measure the changes in root diameter and the root–soil air gap and to calculate the resistance to water flow at the soil–root interface using the above model under different soil water conditions. However, the root system in the root growth chamber, which has limited space, was different from those planted in open fields, thus, the indoor observation results were just used to compare with the outdoor model predictions.

Alongside those planted in open fields, 1-year-old *R. pseudoacacia* plants (~80 cm in height and 0.8 cm in basal diameter) were planted in root growth chambers and placed indoors. The root growth chambers (60 × 60 × 30 cm) were constructed using polymethyl methacrylate, and holes were drilled into the bases of the chambers to allow ventilation and drainage. Air-dried soils collected from the open fields were passed through a soil sieve of 2 mm mesh and carefully packed in the chamber at a similar bulk density to the open fields. Twelve annual *R. pseudoacacia* plants were planted in each chamber, and the chambers were then encased with a specially designed polystyrene box to inhibit algal growth and prevent light exposure to the roots. The polystyrene box was only removed for imaging the root system. Each chamber set an observation profile of the root system; in order to ensure root growth along the observation profile, the boxes were positioned at a slope of ~45° with the observation

profile on the lower side (Marschner and Römhild 1983). The chambers were watered to saturation in early June 2011 and again when the degree of soil water deficit exceeded 0.9.

Images of the root system after watering were captured using a digital camera (Canon EOS 7D, EF-S 15–85 mm f/3.5–5.6 IS USM lens; Canon China Co., Ltd, Shijiazhuang, China) with a pixel resolution of 5184 × 3456. The digital camera was mounted on a tripod in front of the chamber. To capture an image, the polystyrene box encasing the growth chamber was first removed. An L-shaped 15-cm ruler with 1-mm graduations was attached to the left side of the chamber for the purpose of converting the pixel dimensions to millimeters (Futsaether and Oxaal 2002). The digital camera was triggered, and the image of the root system was saved in JPEG format (see Figures S1 and S2 available as Supplementary Data at *Tree Physiology* Online). Microscopic images of the root system were captured using a digital camera connected to a Nikon E200 biological stereomicroscope (Nikon China Co., Ltd, Shanghai, China). The microscope was positioned horizontally against the chamber, allowing an image of the roots within this region to be taken.

After completion of the field measurements, three *R. pseudoacacia* planted in the root growth chamber were selected randomly, and the roots in the observation profile were excavated and imaged using the digital camera. The correlation between the diameter of the actual root system and the value observed from the digital photographs was analysed using the image analysis software ImageJ (version 1.46e, National Institutes of Health, Bethesda, MD, USA, 2012; loading the Object J plug-in). After spatial calibration of the images using the determined relationship between pixel dimensions and millimeters (Futsaether and Oxaal 2002), the changes in root diameter and root–soil air gap were determined by ImageJ software. The root diameter and root–soil air gap at a given location were determined using the microscope images of the roots. The roots were identified by selecting the pixels, and the image was converted to a binary image and skeletonized. The root diameter and root–soil air gap were then determined from the skeletonized image.

Gravimetric soil moisture content was simultaneously determined by collecting soil samples (0.1 kg) at depths ranging from 0 to 10 cm. Fresh soil samples were weighed and transported to the laboratory. The samples were then oven-dried at 105 °C for 24 h and reweighed, allowing the determination of soil moisture content. The soils were subsequently backfilled into the chamber, and soil temperatures at depths of 5, 15 and 25 cm were measured with a standard soil thermometer. Chamber weights were determined with an electronic scale with a precision of 1 g and a 35-kg measurement range.

Statistical analysis

Statistical analyses were performed using the SPSS software package for Windows (Version 18.0, SPSS Inc., Chicago, IL, USA). Prior to analyses, datasets were tested for normal distribution by

the Kolmogorov–Smirnov test and for homogeneity of variances by the Levené test. Data that were not normally distributed were square-root or log transformed. Independent variables were examined for collinearity using a variance inflation factor >10 as the guideline for detection. Mean values of a given variable together with the standard error of the mean were calculated. Stepwise multiple regression analysis was used to test the influence of environmental conditions (soil temperature, soil water potential, soil–root water potential difference, root–air water potential difference and transpiration rate) on the root–soil air gap and water flow resistance at the soil–root interface. The stepwise method was used in regression analyses; a probability value of 0.05 was used to enter variables into the stepwise models, and the removal probability was set at 0.1.

Results

Environmental variables

From June to October 2011 precipitation totalled 407 mm (Figure 1), mean monthly air temperature was 21.3 °C, monthly average temperatures ranged from 13.8 °C (October) to 26.1 °C (July), mean monthly air relative humidity was 69.7%, with monthly average values ranging from 51.4% (June) to 85.7% (August).

Root–soil air gap

We compared the measured root–soil air gap of *R. pseudoacacia* in the root growth chamber with the simulated root–soil air gap in open fields. We also compared the water flow resistances at the soil–root interface of the open fields and the root growth chamber. The average values of parameters used in the water flow model are given in Table 1. The root–soil air gap of *R. pseudoacacia* in open fields from June to October averaged 0.24 mm (Figure 2a, Table 2). After watering, the root–soil air gap

increased gradually from 0.17 mm to ~0.59 mm (Figure 2a, Table 2). The root–soil air gap and water flow resistance at the soil–root interface of *R. pseudoacacia* planted in the root growth chamber were calculated for the period from 27 September to 2 November because complete data for changes in root diameter were available for this time. The average root–soil air gap of *R. pseudoacacia* in this condition was 0.39 mm (Figure 2b, Table 2). The root–soil air gap of *R. pseudoacacia* in open fields exhibits a hysteresis response to soil dehydration and rehydration; a few days after a period of dehydration or rehydration, the root–soil air gap gradually increased or decreased, respectively. As the number of days after watering increased, the mean root–soil air gap first decreased and then gradually increased each month with the exception of August from the fifth to 20th day post watering. The mean gap for July decreased from 0.51 mm on the second day after watering to 0.23 mm on the fifth day, followed by an increase to ~0.33 mm on the 20th day (Figures 2a and 3a, Table 2). The mean gap for August was 0.13 mm on the second day after watering, 0.10 mm on the fifth day and ~0.10 mm on the 20th day (Figures 2a and 3a, Table 2). The diurnal variation

Table 1. The average values of parameters used in the water flow model and numerical simulation. t , Centigrade degree. D_{wv} was calculated from Deng and Xu (1994).

Parameter	Open fields	Root growth chamber
Ψ_s	−0.24	−0.64
Ψ_{ir}	−1.16	−1.50
$\theta(c)$	0.86	0.55
D_{wv}	0.0005	0.04
J_v	0.00001	0.000005
R	8.314472×10^{-6}	
T	$t + 273.15$	
V_w	0.018×10^{-6}	
D_{wv}	Deng and Xu (1994)	

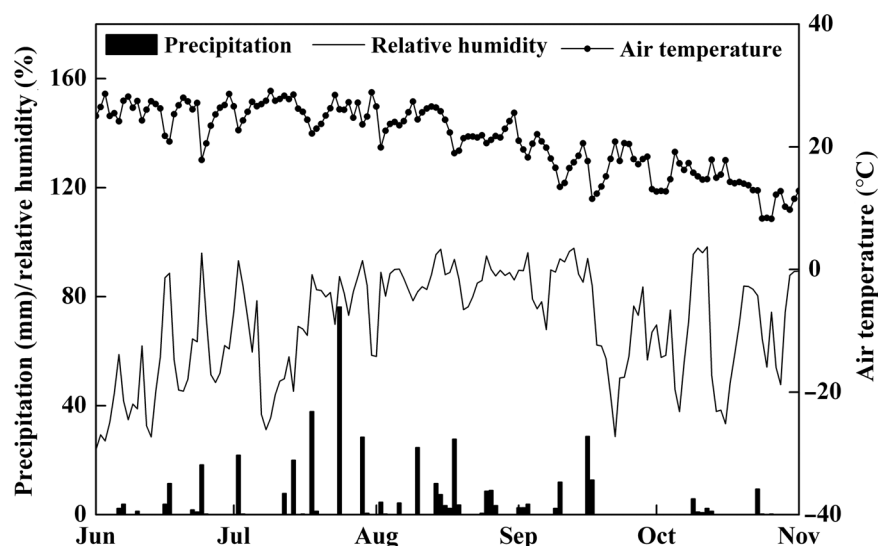


Figure 1. Seasonal patterns of daily precipitation, daily air temperature and daily air relative humidity from June to October 2011.

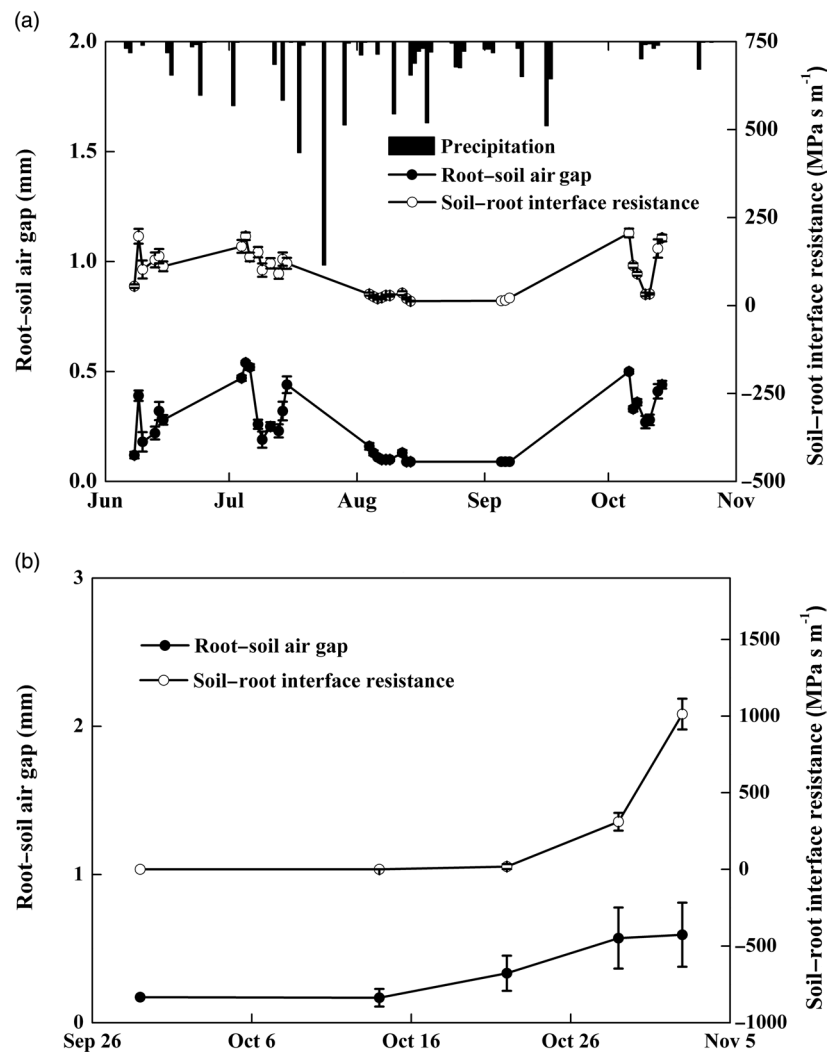


Figure 2. Root-soil air gap and water flow resistance at the soil-root interface of *R. pseudoacacia* plants in different months. (a) Open fields and (b) root growth chamber.

Table 2. Root-soil air gap and water flow resistance at the soil-root interface of *R. pseudoacacia* plants. A: open fields; B: root growth chamber.

	Root-soil air gap (mm)	Water flow resistance at the soil-root interface (MPa s m ⁻¹)
A Month		
June	0.25 ± 0.04	123 ± 19.1
July	0.36 ± 0.05	135 ± 11.1
August	0.11 ± 0.01	24.9 ± 2.35
October	0.37 ± 0.03	118 ± 27.0
July		
Second day post watering	0.51 ± 0.02	168 ± 20.2
Fifth day post watering	0.23 ± 0.04	119 ± 23.6
20th day post watering	0.33 ± 0.07	134 ± 19.7
August		
Second day post watering	0.13 ± 0.02	27.2 ± 4.30
Fifth day post watering	0.10 ± 0.01	26.4 ± 3.34
20th day post watering	0.10 ± 0.01	22.5 ± 6.64
B 27 September	0.17 ± 0.04	0.11 ± 0.08
2 November	0.59 ± 0.22	1012 ± 100

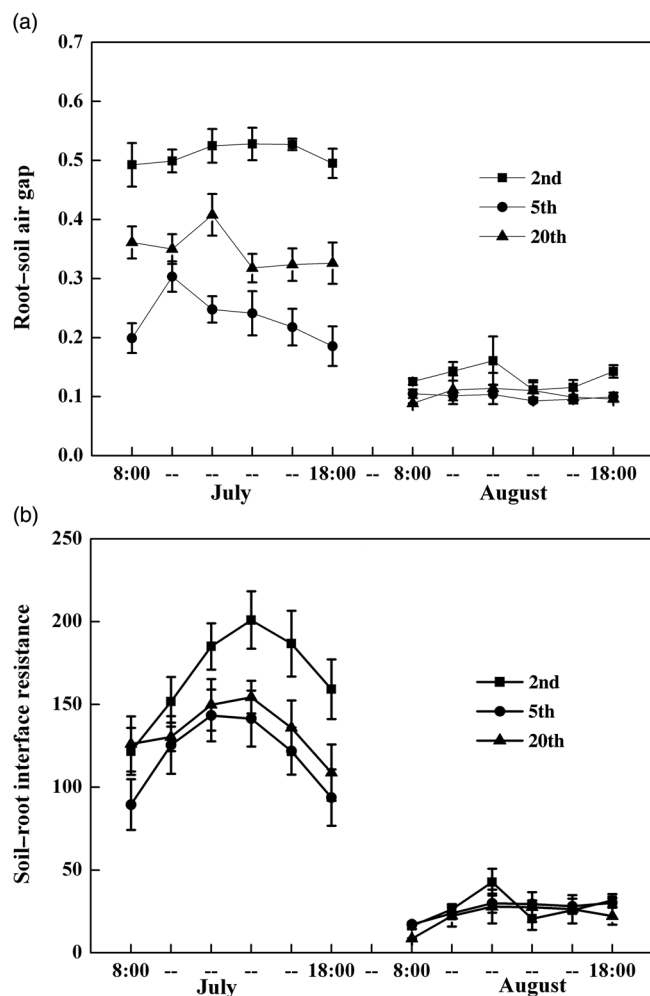


Figure 3. Daily variations in root-soil air gap and water flow resistance at the soil-root interface of *R. pseudoacacia* planted in open fields. (a) Root-soil air gap and (b) water flow resistance at the soil-root interface. 2nd, 5th and 20th refer to number of days post watering.

in the root-soil air gap in open fields was described by a single-peak curve with minima in the morning and evening and a maximum at noon (Figure 3a). With the exception of the root-soil air gap in August on the second day, the minimum values appeared at ~8:00 and 16:00 h, while the maximum value appeared at ~12:00–14:00 h (Figure 3a).

The response of the root-soil air gap of *R. pseudoacacia* in open fields to soil dehydration indicated that the initial gap increased with decreasing soil water potential, and at low values of soil water potential, the root-soil air gap reached a high-value plateau (Figure 4). At the initial soil water potential of -0.002 MPa, the root-soil air gap was ~ 0.09 mm. When soil water potential decreased to -0.75 MPa, the air gap increased to 0.54 mm (Figure 4). With increasing soil temperature, the root-soil air gap in open fields decreased slightly. The average root-soil air gap decreased from 0.36 mm at 15.0 °C to 0.23 mm at 24.0 °C (Figure 5). Stepwise regression analysis on *R. pseudoacacia* planted in both open fields and in the root growth chamber

revealed that soil temperature and soil water potential could best explain the variation in the root-soil air gap (Table 3).

Resistance to flow at the soil-root interface

The average water flow resistance at the soil-root interface of *R. pseudoacacia* planted in open fields from June to October was 88.4 MPa s m⁻¹ (Figure 2a, Table 2). After watering, the mean soil-root interface water flow resistance decreased and then increased with the exception of August from the fifth to 20th day post watering. The mean resistance for July decreased from 168 MPa s m⁻¹ on the second day after watering to 119 MPa s m⁻¹ on the fifth day followed by an increase to ~ 134 MPa s m⁻¹ on the 20th day (Figures 2a and 3b, Table 2). The mean resistance for August was 27.2 MPa s m⁻¹ on the second day, 26.4 MPa s m⁻¹ on the fifth day and ~ 22.5 MPa s m⁻¹ on the 20th day (Figures 2a and 3b, Table 2). The average water flow resistance at the soil-root interface of *R. pseudoacacia* grown in the root growth chamber was 174 MPa s m⁻¹, and the average resistance increased gradually with increasing time after watering (Figure 2b, Table 2). Due to the influence of environmental factors, especially soil water potential, along with the influences of plant transpiration pull and water potential gradient, the water flow resistance at the soil-root interface showed a regular pattern in daily variation. The diurnal variation in open fields was described by a single-peak curve with a maximum at noon and minima in the morning and afternoon. With the exception of August on the second day after watering, the minimum values of water flow resistance at the soil-root interface appeared at ~08:00 and 16:00 h, while the maximum value appeared at ~12:00–14:00 h (Figure 3b). Similar to the root-soil air gap, the response of the water flow resistance to soil dehydration and rehydration exhibits hysteresis.

The trends in water flow resistance at the soil-root interface of *R. pseudoacacia* planted in open fields were similar to those of the root-soil air gap (Figure 4). As the soil water potential decreased, the growing roots shrank, the root-soil air gap appeared and the resistance to water flow at the soil-root interface increased, reaching a plateau at a high value corresponding to low soil water potential (Figure 4). At the initial soil water potential of -0.002 MPa, the water flow resistance at the soil-root interface of *R. pseudoacacia* in open fields was 19.9 MPa s m⁻¹. When the soil water potential decreased to -0.75 MPa, the resistance value increased to 197 MPa s m⁻¹ (Figure 4). With increasing soil temperature, the resistance to water flow at the soil-root interface in open fields decreased slightly; the average resistance increased from 103 MPa s m⁻¹ at 15.0 °C to 80.1 MPa s m⁻¹ at 24.0 °C (Figure 5). Stepwise regression analysis of *R. pseudoacacia* planted in open fields and in the root growth chamber demonstrated that a combination of soil water potential, soil temperature, root-air water potential difference and soil-root water potential difference could best predict the resistance to water flow at the soil-root interface (Table 3).

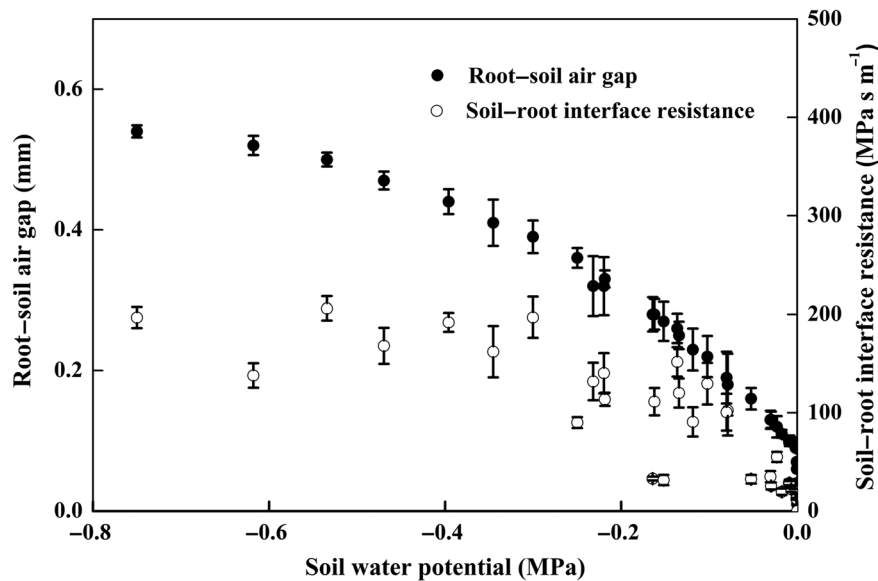


Figure 4. Root-soil air gap and water flow resistance at the soil-root interface of *R. pseudoacacia* planted in open fields under different soil water potentials.

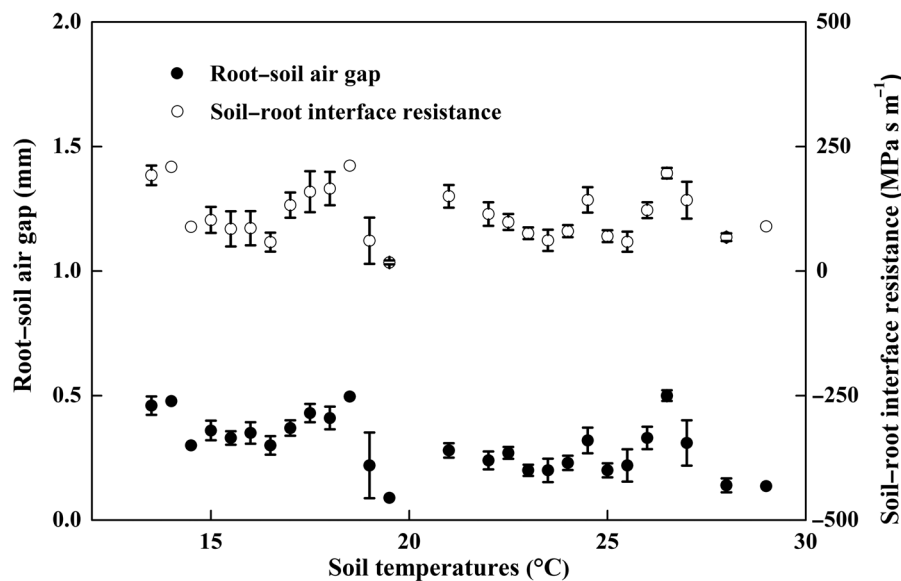


Figure 5. Root-soil air gap and water flow resistance at the soil-root interface of *R. pseudoacacia* planted in open fields under different soil temperatures.

Relationship between the water flow resistance at the soil-root interface and the root-soil air gap

In this study, the resistance to water flow at the soil-root interface of *R. pseudoacacia* planted in open fields and in the root growth chamber was strongly correlated with the root-soil air gap (Figure 6). A linear and power function regression model was fitted to describe the relationship between them, while using a linear equation, the resistance to water flow at the soil-root interface is negative when the root-soil air gap reaches zero; the relationship is described by a typical power function where resistance to water flow at the soil-root interface increases sharply with increasing root-soil air gap. Accordingly,

as soil water potential decreases, the root-soil air gap and water flow resistance at the soil-root interface decrease.

Sensitivity analysis

The parameters included in the sensitivity test were transpiration rate, soil temperature, soil water potential, air temperature and air water potential. One parameter at a time, the values of the above parameters were increased or decreased by 5% (0.5 °C), 10% (1 °C) and 20% (2 °C) in order to monitor the effects on root-soil air gap and resistance to water flow at the soil-root interface (Wang et al. 2008). As depicted in Figure 7, the modelled root-soil air gap was less sensitive to transpiration rate,

Table 3. Stepwise regression analysis for the relationship between root–soil air gap, water flow resistance at the soil–root interface and environmental conditions.

	Root–soil air gap		Water flow resistance at the soil–interface	
	r^1	P^2	r^1	P^2
Soil water potential	−0.593	0.001	−0.317	0.045
Soil temperature	−0.257	0.114	−0.303	0.038
Root–air water potential difference	–	–	1.029	0.000
Soil–root water potential difference	–	–	−0.470	0.079
Overall model ³		0.002		0.000

¹The partial regression coefficient (r) value in the stepwise regression analysis.

²The partial P -value for the specific factor.

³The statistics for the regression model combining all factors that were significant at $P < 0.01$.

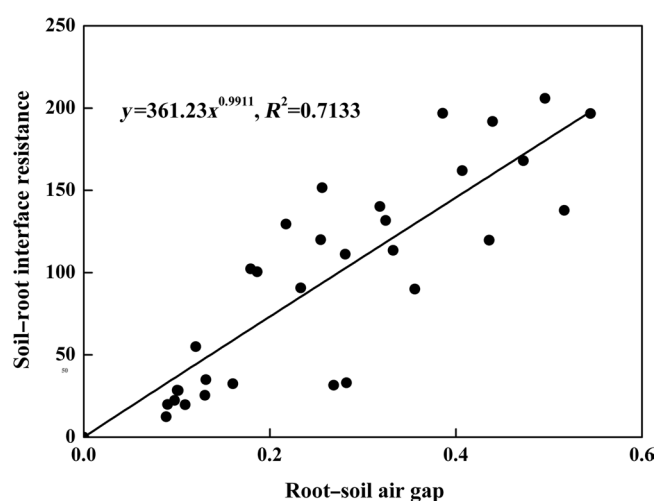


Figure 6. Relationship between root–soil air gap and water flow resistance at the soil–root interface of *R. pseudoacacia* planted in open fields and in the root growth chamber.

soil temperature, air temperature and air water potential and more sensitive to soil water potential. Soil temperature, soil water potential, air temperature and air water potential all had observable effects on the root–soil air gap. For the modelled resistance to water flow at the soil–root interface, transpiration rate had the smallest effect, whereas it exhibited a highly linear sensitivity to soil water potential and air water potential. The water flow resistance model exhibited a negative linear response to air temperature to some degree along with a slightly positive linear response to soil temperature.

Discussion

In this study, the models determined for root–soil air gap and resistance to water flow at the soil–root interface of *R. pseudoacacia* when water is transported across the soil–root interface under different soil water conditions are similar to those developed by

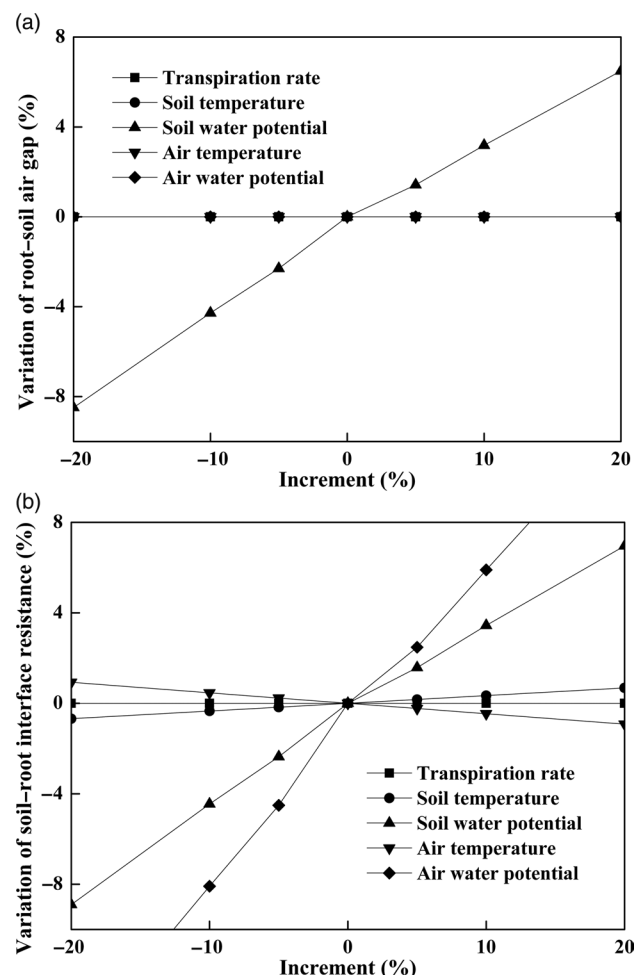


Figure 7. Plot of sensitivity of estimated root–soil air gap and water flow resistance at the soil–root interface to the model parameters.

Nobel and Cui (1992a), who described water vapour diffusion across a root–soil air gap under isothermal conditions based on Fick's first law. Their study found that the soil, root–soil air gap and root could limit water exchange between the soil and root at different soil water potentials. Advances in imaging techniques allow the in situ visualization of changes in root diameter with high spatial and temporal resolution (Carminati et al. 2010). Our study employed a digital camera and a stereomicroscope to image changes in *R. pseudoacacia* root diameters. Similarly, to investigate rhizosphere properties, Carminati et al. (2010) utilized neutron radiography to image water content distributions in soil samples planted with lupins during drying and subsequent rewetting.

Soil shrinkage is related to the changes in soil volume with soil water content in the process of soil drying and wetting (Lu and Shao 2003). Soils with higher clay contents such as montmorillonite, nontronite, beidellite and vermiculite exhibit more soil shrinkage (Huang 2000). However, soils in the area relevant to this study are generally classified as mountainous cinnamon soils (Ustalf); thus, soil shrinkage can be disregarded. Losses in soil–root contact have been observed in water-deficient roots,

primarily due to root shrinkage (Huck et al. 1970, Huang and Kang 1997). As the root shrinks, it can no longer fill the root channel and a gap may form around it on all sides or, more likely, remain in contact on one side (Tinker 1976, Faiz and Weatherley 1982, Nye 1994). The root–soil air gap of *R. pseudoacacia* planted in open fields and in the root growth chamber demonstrated progressively greater root shrinkage and loss of root–soil contact as soil water potential decreased. The difference between the calculated and measured values of the root–soil air gap can be attributed to the stand age of *R. pseudoacacia* and the space limitations of the root growth chamber. Similarly, the roots of *Agave deserti* Engelm. shrank by 34% in containers and 40% in the field (North and Nobel 1997b) during drying. This result might partly reflect the greater average root age in the field; cortical cells shrink and die during ageing, a process that is hastened by drought (North and Nobel 1995). Changes in the root–soil air gap have also been observed in other species. The root–soil air gap of *Helianthus annuus* L. increased to 0.08 mm when the soil water potential decreased from -0.2 to -1.2 MPa (Faiz and Weatherley 1982). This represents a significantly smaller change than those observed in this study for *R. pseudoacacia*. The root–soil air gap of desert succulents increased up to 0.38 mm after 30 days of soil drying (Nobel and Cui 1992a), which is similar to the present results for *R. pseudoacacia* planted in the root growth chamber. In contrast, Taylor and Willatt (1983) reported that the value of the root–soil air gap of *Glycine max* Willd., which was slightly higher than that in this study, could increase up to 3 mm as the water potential decreased from 0.0 to -1.3 MPa.

The soil water potential of *R. pseudoacacia* in open fields first increased and then decreased after watering; as a result, the root–soil air gap first decreased and then increased after watering because the gap increases with decreasing soil water potential (Figures 2a and 3a). This result might indicate that the root–soil air gap was affected by environmental factors; for example, precipitation was one of the most important factors affecting soil water potential. We found that the response of the root–soil air gap to soil dehydration and rehydration exhibits a hysteresis phenomenon. Nobel and Cui (1992a) and Xu et al. (1999) obtained a similar result. This response might be related to soil texture and plant species in unsaturated soil. As soil is dehydrated or water is taken up by plant roots, the water content at the soil–root interface decreases, resulting in a corresponding increase in the root–soil air gap. The change in the soil–root air gap depends on the rate of water uptake by the root along with the ability of the soil to replenish the water lost around that root (Xu et al. 1999). The stepwise regression analysis revealed that soil temperature and soil water potential could best predict the variation in the root–soil air gap (Table 3). These findings are comparable to numerous field studies documenting strong relationships between soil temperature, soil water potential and root–soil air gap (Guo et al. 1996, Huang and Kang 1997, Kang and Zhang 1997). Kang and Zhang (1997) found that

temperature had no effect on maize and sunflower roots when soil water potential exceeded -1.0 MPa; below this potential value, root shrinkage increased with increasing temperature. Our results showed that the root–soil air gap and water flow resistance at the soil–root interface of *R. pseudoacacia* in open fields increased with decreasing soil water potential (Figure 3) and decreased with increasing soil temperature (Figure 5).

The shrinkage and subsequent rehydration of plant roots growing in soil cause significant variations in the resistance to water flow at the soil–root interface (Faiz and Weatherley 1982, Taylor and Willatt 1983, Nye 1994). Decreases in soil–root contact may contribute to the increased soil–root interface water flow resistance (Huck et al. 1970, Faiz and Weatherley 1982, Taylor and Willatt 1983). However, Faiz and Weatherley (1982) suggested that root contraction is not sufficient to generate an increase in the interfacial resistance. In the present study, the water flow resistance at the soil–root interface of *R. pseudoacacia* increased with an increasing root–soil air gap (Figure 6). The present study indicates that the combination of soil water potential, soil temperature, root–air water potential difference and soil–root water potential difference can best predict the resistance to water flow at the soil–root interface. Similarly, Nobel and Cui (1992a) and Braud et al. (2005) reported that water movement into roots depends on the water potential difference and soil–root resistance from the bulk soil to the root xylem. Taylor and Willatt (1983), however, found that root shrinkage of plants growing in moist soil generates negligible increases in the root–soil contact resistance. In addition, other studies indicate that the root–soil air gap is unimportant (Cowan and Milthorpe 1968), and that vapour-phase transport is negligible until soils become too dry to support reasonable plant growth (Rose 1963, Newman 1969b). In very dry conditions, vapour transfer across the root–soil air gap can contribute significantly to the total inflow of water (Nye 1994).

Conclusion

The root–soil air gap of *R. pseudoacacia* in open fields and in the root growth chamber demonstrated progressively greater root shrinkage and loss of root–soil contact with decreasing soil water potential. The resistance to water flow at the soil–root interface of *R. pseudoacacia* increased with decreasing soil water potential. Stepwise regression analysis indicated that soil temperature and soil water potential could be best used to predict the variation in root–soil air gap. A combination of soil water potential, soil temperature, root–air water potential difference and soil–root water potential difference was found to best predict the water flow resistance at the soil–root interface.

Supplementary data

Supplementary data for this article are available at *Tree Physiology* Online.

Conflict of interest

None declared.

Funding

This research was supported by the National Natural Science Foundation of China (31000251), 'Strategic Priority Research Program' of the Chinese Academy of Sciences (XDA05060600), Youth Innovation Promotion Association, CAS (2014083) and Special Fund for Forest Scientific Research in the Public Welfare (201404214).

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